

Emanuele Rossi

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Researcher working on machine learning for animal communication, part of a broader interest in leveraging ML to recover structure in complex systems: previously ML on graphs (Temporal Graph Networks, SIGN) at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

Google Scholar metrics: 4,444 citations; h-index 15 (August 2026).

Education

Imperial College London <i>Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein</i> Research focused on graph machine learning methods for large real-world systems	<i>Oct 2020 – Feb 2024</i>
University of Cambridge <i>Master in Advanced Computer Science, Distinction (4.0 GPA)</i>	<i>Oct 2018 – Jun 2019</i>
Imperial College London <i>Bachelor in Computer Science, First Class Honors (4.0 GPA)</i>	<i>Oct 2015 – Jun 2018</i>

Experience

Postdoctoral Researcher <i>GLADIA Research Group</i> 🔗 , Sapienza University of Rome – With Prof. E. Rodolà	<i>Rome (remote)</i> <i>Oct 2025 – Present</i>
Developing novel machine-learning methods to deepen our understanding of non-human communication.	
Machine Learning Researcher <i>VantAI</i>	<i>NYC (remote)</i> <i>Jan 2024 – Aug 2025</i>
Co-led development of Neo-1 🔗, a state-of-the-art all-atom latent diffusion model for multimodal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.	
Machine Learning Researcher <i>Twitter</i>	<i>London</i> <i>Jun 2019 – Feb 2023</i>
Developed graph ML models for Twitter’s social graph and deployed a recommendation model that generated 1.2M additional daily follows	
Machine Learning Researcher <i>Fabula AI</i>	<i>London</i> <i>Mar 2019 – Jun 2019</i>
Developed graph ML models for fake news detection as part of Fabula AI’s small research team before Twitter acquired the company 🔗	
Software Engineering Intern <i>Google</i>	<i>California</i> <i>Jul 2017 – Oct 2017</i>

Selected Publications

[Full list](#)

Animal Communication & Audio	
<i>D. Marincione, D. Crisostomi, R. Dessì, E. Rodolà, and E. Rossi. Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models</i> 🔗. <i>NeurIPS Workshop on AI for Animal Communication 2025, Spotlight (top 15%)</i>. <i>F. Mustun, C. Semenzin, R. Dessì, P. R. Guerrero, P. Orhan, A. Emanuelli, E. Rossi et al. OpenWhistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. Under Review 2026.</i> <i>E. Rossi and E. Rodolà. Communicating Sound Through Natural Language</i> 🔗. <i>arXiv 2026</i>.	
Graph Learning	
<i>E. Rossi et al. Temporal Graph Networks for Deep Learning on Dynamic Graphs</i> 🔗. <i>ICML GRL Workshop 2020. 1,700+ citations and 1,000+ GitHub stars</i> 🔗. <i>E. Rossi*, F. Frasca* et al. SIGN: Scalable Inception Graph Neural Networks</i> 🔗. <i>ICML GRL Workshop 2020. 600+ citations and used in production at Airbnb</i> 🔗. <i>B. Chamberlain, S. Shirobokov, E. Rossi et al. Graph Neural Networks for Link Prediction with Subgraph Sketching</i> 🔗. <i>ICLR 2023, Oral (top 5%). 200+ citations.</i>	

Leadership

Co-Founder, LeadTheFuture (Forbes article 🔗)	<i>Jun 2018 – Mar 2023</i>
Built a non-profit mentoring community with more than 200 mentors and 500 selected mentees	

Supervision and Mentoring

- **Davide Marincione** – PhD Student @ Sapienza University of Rome 05/2025 – 03/2026
Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models
- **Simone Antonelli** – ML Research Intern @ Amboss Technologies 10/2024 – 07/2026
Predicting Link Closure in the Lightning Network with Temporal Graph Networks
- **Marco Pegoraro** – ML Research Intern @ VantAI 10/2024 – 04/2025
Neo 1: a unified model for all-atom structure prediction and generation of all biomolecules
- **Harrison Rush** – ML Engineer @ Amboss Technologies 08/2024 – Present
MPFlow: Learning Budgeted Max-Flow Optimization on the Lightning Network with Deep Graph Reinforcement Learning
- **Thomas Castiglione** – ML Researcher @ VantAI 03/2024 – 04/2025
Neo 1: a unified model for all-atom structure prediction and generation of all biomolecules
- **Andy Huang** – PhD Student @ Mila 03/2024 – 04/2025
TGB: Temporal Graph Benchmark for Machine Learning on Temporal Graphs
UTG: Towards a Unified View of Snapshot and Event Based Models for Temporal Graphs
- **Vincent Davis** – ML Engineer @ Amboss Technologies 11/2023 – 05/2024
Channel Balance Interpolation in the Lightning Network via Machine Learning

Research Grants & Fellowships

- **ISCRA-C Research Computing Grant** – CINECA 02/07/2026
DOLPH-ID: Dolphin Emitter Identification from Whistle Acoustics ≈€20,000 equivalent
Allocation of 8,500 GPU-hours on Leonardo's NVIDIA A100 64 GB accelerators. 9 months
- **Research Fellowship** – Sapienza University of Rome 01/04/2026
For postdoctoral research on machine learning for non-human communication. €36,000
12 months
- **Google Cloud Research Credits** – Google 2022, 2023
GCP credits for PhD research. Received in two consecutive years. \$1,000/year

Awards

- **Olav Beckmann Project Prize** – Imperial College London 2017
For outstanding second-year project work.
- **Engineering Dean's List** – Imperial College London 2016, 2017
Top 10% of cohort. Received in two consecutive years.
- **G-Research Prize (Gloucester Research Ltd Prize at the time of award)** – Imperial College London 2016
Awarded annually to up to 10 non-final-year Computer Science students for academic excellence.
- **President's Undergraduate Scholarship** – Imperial College London 22/05/2015
Imperial's most prestigious undergraduate distinction, awarded to fewer than 5% of the incoming cohort for academic excellence and potential. Funding: £3,000.

Invited Talks

- *Graph Machine Learning for Dynamic Financial Networks: From Foundations to the Lightning Network* 10/07/2026
[Swissquote Workshop on Graph Learning in Financial Networks, EPFL \[Slides \(HTML\)\] \[Slides \(PDF\)\]](#)
- *Machine Learning on Temporal Graphs: From Foundations to Emerging Frontiers* 19/05/2026
Linguae-ML Seminar, Ecole Normale Supérieure (Paris) [\[Slides \(HTML\)\] \[Slides \(PDF\)\]](#)
- *Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models* 30/01/2026
Earth Species Project Community Deep Dive [\[Slides \(PDF\)\] \[Video\]](#)
- *A Primer on Graph Neural Networks* 22/11/2023
COLT Seminar, University of Pompeu Fabra, Barcelona [\[Slides \(PDF\)\]](#)
- *Edge Directionality Improves Learning on Heterophilic Graphs* 14/09/2023
[Temporal Graph Reading Group \[Slides \(PDF\)\] \[Video\]](#)
- *Edge Directionality Improves Learning on Heterophilic Graphs* 07/09/2023
[Explainability and Applicability of GNNs Workshop, Kassel](#)
- *Improving Machine Learning at Twitter using Graphs* 28/09/2022
[Privacy Week \[Video\]](#)

◦ <i>Learning on Graphs with Missing Node Features</i> TigerGraph Reading Group	09/09/2022
◦ <i>Learning on Graphs with Missing Node Features</i> Learning on Graphs and Geometry Reading Group [Slides (PDF)] [Video]	01/03/2022
◦ <i>Learning on Graphs with Missing Node Features</i> Machine Learning Research Group, University of Oxford	24/02/2022
◦ <i>Learning on Graphs with Missing Node Features</i> University of Cambridge AI Research Group Talks	23/11/2021
◦ <i>Graph Neural Networks with Almost No Features</i> Toronto Machine Learning Summit	17/11/2021
◦ <i>Machine Learning on Dynamic Graphs: Temporal Graph Networks</i> MLOps World 2021	16/06/2021
◦ <i>Machine Learning on Dynamic Graphs: Temporal Graph Networks</i> Intel ML Seminar [Slides (PDF)]	26/04/2021
◦ <i>Machine Learning on Dynamic Graphs: Temporal Graph Networks</i> GNNSys'21 Workshop	09/04/2021
◦ <i>Dynamic Graphs and Temporal Graph Networks</i> University of Liverpool Networks and Distributed Computing Series [Slides (PDF)]	22/01/2021
◦ <i>TGN: Temporal Graph Networks</i> AISC [Slides (PDF)] [Video]	21/07/2020

Academic Service

◦ Post-AGI Science and Society Workshop – ICLR 2026 PC Member	04/2026
◦ Temporal Graph Learning Workshop – NeurIPS 2023 Organizer	12/2023
◦ Temporal Graph Learning Workshop – NeurIPS 2022 Panel	12/2022
◦ NeurReps Workshop – Symmetry and Geometry in Neural Representations – NeurIPS 2022 PC Member	12/2022
◦ GCLR 22 – 2nd Workshop on Graphs and more complex structures for learning and reasoning – AAAI 2022 PC Member	02/2022
◦ GReS – Workshop on Graph Neural Networks for Recommendation and Search – ACM RecSys 2021 PC Member	10/2021
◦ GNNSys'21 – Workshop on Graph Neural Networks and Systems – MLSys 2021 PC Member	04/2021
◦ AAAI-21 GCLR – AAAI 2021 PC Member	02/2021

Full Publication List

Under Review

- F. Mustun, C. Semenzin, R. Dessì, P. R. Guerrero, P. Orhan, A. Emanuelli, **E. Rossi**, Y. Lakretz, G. G. de Polavieja and G. Sumbre. 2026. OpenWhistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. *Under Review*. (0).
- H. Rush, V. Davis, S. Antonelli, V. Singh, J. Shrader and **E. Rossi**. 2026. MPFlow: Learning Budgeted Max-Flow Optimization on the Lightning Network with Deep Graph Reinforcement Learning. *arXiv*. (0).
- **E. Rossi** and E. Rodolà. 2026. Communicating Sound Through Natural Language. *arXiv*. (0).
- S. Antonelli, V. Davis, H. Rush, A. Potdevin, J. Shrader, V. Singh and **E. Rossi**. 2026. Predicting Channel Closures in the Lightning Network with Machine Learning. *arXiv*. (0).
- D. Kovtun, M. Akdel, A. Goncarenco, G. Zhou, G. Holt, D. Baugher, D. Lin, Y. Adeshina, T. Castiglione, X. Wang, C. Marquet, M. McPartlon, T. Geffner, **E. Rossi**, G. Corso, H. Stärk, Z. Carpenter, E. Kucukbenli, M. Bronstein and L. Naef. 2024. PINDER: The Protein Interaction Dataset and Evaluation Resource. *bioRxiv*. (37).

Peer-reviewed journal articles

- D. Malitesta, **E. Rossi**, C. Pomo, T. Di Noia and F. D. Malliaros. 2026. Training-Free Graph-Based Imputation of Missing Modalities in Multimodal Recommendation. *IEEE Transactions on Knowledge and Data Engineering*. (2).
- Y. Aluma, Z. Baron, R. Barrett, C. Baumgartner, G. Beguš, S. Bhattacharya, M. M. Bronstein, S. Dahan, O. Davis, S. de Haas, J. Defoe, J. DelPreto, R. Dessì, R. Diamant, J. Gatesy, K. George, S. Gero, D. Gibbons, D. Gibbons, S. Gil, S. Goldwasser, D. F. Gruber, O. Harve, A. Hernandez, M. Ishay, N. Jadhav, L. KC, A. Kenny, A. Leitao, M. Lucas, A. Maalouf, P. Malkin, Y. Mevorach, S. Pagani, O. Paradise, G. Petri, S. Poetto, **E. Rossi**, D. Rus, M. Salino-Hugg, A. Santoro, P. Sharma, D. Tchernov, A. Torralba, P. Tønnesen, D. M. Vogt and R. J. Wood. 2026. Description of a collaborative sperm whale birth and shifts in coda vocal styles during key events. *Scientific Reports*. (1).
- V. Singh, M. Khanzadeh, V. Davis, H. Rush, **E. Rossi**, J. Shrader and P. Lio. 2025. Bayesian Binary Search. *Algorithms*. (6).

Peer-reviewed conference proceedings

- V. Davis, V. Singh and **E. Rossi**. 2025. Channel Balance Interpolation in the Lightning Network via Machine Learning. *IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*. (4).
- J. Gastinger, S. Huang, M. Galkin, E. Loghmani, A. Parviz, F. Poursafaei, J. Danovitch, **E. Rossi**, I. Koutis, H. Stuckenschmidt, R. Rabbany and G. Rabusseau. 2024. TGB 2.0: A Benchmark for Learning on Temporal Knowledge Graphs and Heterogeneous Graphs. *Advances in Neural Information Processing Systems*. (47).
- D. Malitesta, **E. Rossi**, C. Pomo, T. Di Noia and F. D. Malliaros. 2024. Do We Really Need to Drop Items with Missing Modalities in Multimodal Recommendation?. *Proceedings of the 33rd ACM International Conference on Information*. (50).
- S. Huang, F. Poursafaei, R. Rabbany, G. Rabusseau and **E. Rossi**. 2024. UTG: Towards a Unified View of Snapshot and Event Based Models for Temporal Graphs. *Learning on Graphs Conference (LoG)*. (20).
- S. Huang, F. Poursafaei, J. Danovitch, M. Fey, W. Hu, **E. Rossi**, J. Leskovec, M. M. Bronstein, G. Rabusseau and R. Rabbany. 2023. Temporal Graph Benchmark for Machine Learning on Temporal Graphs. *Advances in Neural Information Processing Systems*. (273).
- **E. Rossi**, B. Charpentier, F. Di Giovanni, F. Frasca, S. Günnemann and M. M. Bronstein. 2023. Edge Directionality Improves Learning on Heterophilic Graphs. *Learning on Graphs Conference (LoG)*. (202).
- B. Chamberlain, S. Shirobokov, **E. Rossi**, F. Frasca, T. Markovich, N. Hammerla, M. M. Bronstein and M. Hansmire. 2023. Graph Neural Networks for Link Prediction with Subgraph Sketching. *International Conference on Learning Representations (ICLR)*. **Oral (top 5%)**. (230).
- M. Mutti, R. De Santi, **E. Rossi**, J. F. Calderon, M. M. Bronstein and M. Restelli. 2022. Provably Efficient Causal Model-Based Reinforcement Learning for Systematic Generalization. *Proceedings of the AAAI Conference on Artificial Intelligence*. (36).
- **E. Rossi**, F. Monti, Y. Leng, M. M. Bronstein and X. Dong. 2022. Learning to Infer Structures of Network Games. *Proceedings of the 39th International Conference on Machine Learning, ICML*. (13).
- **E. Rossi**, H. Kenlay, M. I. Gorinova, B. Chamberlain, X. Dong and M. M. Bronstein. 2022. On the Unreasonable Effectiveness of Feature propagation in Learning on Graphs with Missing Node Features. *Learning on Graphs Conference (LoG)*. (196).
- B. Chamberlain, J. Rowbottom, M. I. Gorinova, S. D. Webb, **E. Rossi** and M. M. Bronstein. 2021. GRAND: Graph Neural Diffusion. *Proceedings of the 38th International Conference on Machine Learning, ICML*. (544).
- B. Chamberlain, **E. Rossi**, D. Shiebler, S. Sedhain and M. Bronstein. 2020. Tuning Word2vec for Large Scale Recommendation Systems. *RecSys - 14th ACM Conference on Recommender Systems*. (38).

Peer-reviewed workshop proceedings

- D. Marincione, D. Crisostomi, R. Dessì, E. Rodolà and **E. Rossi**. 2025. Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models. *NeurIPS Workshop on AI for Animal Communication*. **Spotlight (top 15%)**. (3).
- J. Durairaj, Y. Adeshina, Z. Cao, X. Zhang, V. Oleinikovas, T. Duignan, Z. McClure, X. Robin, G. Studer, D. Kovtun, **E. Rossi**, G. Zhou, S. Veccham, C. Isert, Y. Peng, P. Sundareson, M. Akdel, G. Corso, H. Stärk, G. Tauriello, Z. Carpenter, M. Bronstein, E. Kucukbenli, T. Schwede and L. Naef. 2024. PLINDER: The Protein-Ligand Interactions Dataset and Evaluation Resource. *ICML ML for Life and Material Science Workshop*. (95).
- **E. Rossi**, B. Chamberlain, F. Frasca, D. Eynard, F. Monti and M. Bronstein. 2020. Temporal Graph Networks for Deep Learning on Dynamic Graphs. *ICML Workshop on Graph Representation Learning*. (1747).

- **E. Rossi**, F. Frasca, B. Chamberlain, D. Eynard, M. Bronstein and F. Monti. 2020. SIGN: Scalable Inception Graph Neural Networks. *ICML Workshop on Graph Representation Learning*. (600).
- **E. Rossi**, F. Monti, M. Bronstein and P. Lio. 2019. ncRNA Classification with Graph Convolutional Networks. *KDD Workshop on Deep Learning on Graphs*. (40).